

# INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

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**LABORATORY ASSIGNMENT REPORT**

**ASSIGNMENT 5: TEST SETUP AND TEARDOWN**

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| <b>Department</b>       | : | CSE (AI & ML)         |
| <b>Course</b>           | : | Python for Automation |
| <b>Academic Session</b> | : | 2023-2027             |
| <b>Submission Date</b>  | : | 28 Sept, 2026         |

## 1. MODULE NAME & EXPERIMENT TITLE

**Module Name:** Test Setup and Teardown

**Experiment Title:** Automation of Test Lifecycle Hooks (Test Setup & Test Teardown) for Web Application Login and Resource Cleanup in Robot Framework

## 2. PROBLEM STATEMENT

Design and implement an automated test suite using Robot Framework that incorporates standard setup and teardown routines:

- Set up a test suite with a common test setup that opens a browser and logs in before each test case.
- Define a test teardown that logs out or cleans up resources (closing the browser instance) after each test case.

## 3. OBJECTIVE

- To implement reusable lifecycle management keywords in Robot Framework using Test Setup and Test Teardown.
- To ensure test independence and clean environment states by automatically initializing preconditions (launching browser, logging in) and postconditions (logging out, closing browser).
- To structure test cases efficiently by removing duplicate setup and cleanup steps from individual test bodies.

## 4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language:** Python 3.11.x
- **Automation Framework:** Robot Framework 6.1
- **Library Used:** SeleniumLibrary (Web browser automation)
- **Target Test Web Application:** SauceDemo / Demo Login Portal (<https://www.saucedemo.com>)
- **IDE / Development Environment:** Visual Studio Code (VS Code)
- **Core Concepts:** Preconditions (Test Setup), Postconditions (Test Teardown), Suite Hooks, Clean Resource Teardown, Browser Session Handling.

## 5. SOURCE CODE / IMPLEMENTATION STEPS

Test Suite Script (assignment5\_tests.robot)

```
assignment5_tests.robot - Test Setup & Teardown Implementation Robot Framework

1  *** Settings ***

2  Documentation Assignment 5 - Common Test Setup and Teardown Automation

3  Library SeleniumLibrary

4  Test Setup Open Browser And Login

5  Test Teardown Logout And Close Browser

6
```

## 6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The test suite was executed in VS Code terminal via `robot assignment5_tests.robot`. Robot Framework automatically triggered 'Open Browser And Login' before each test and 'Logout And Close Browser' after each test completed.

**Screenshot 1: Terminal Execution Log displaying Setup, Test Runs, and Teardown**

```
PS C:\Users\Ashmita\Automation_Lab> robot assignment5_tests.robot
===== Assignment5 Tests =====
===== [SETUP] Opening browser and logging in standard_user... TC_01: Verify Product Dashboard Header After Login | PASS
| [TEARDOWN] Logging out standard_user and closing browser session... =====
===== [SETUP] Opening browser and logging in standard_user... TC_02: Verify Item Addition To Cart | PASS | [TEARDOWN] Logging out standard_user
and closing browser session... =====
===== Assignment5 Tests | PASS | 2 tests, 2 passed, 0 failed =====
===== Output: =====
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html Report:
C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

**Screenshot 2: Detailed HTML Test Log Breakdown (log.html)**

### Robot Framework Test Execution Log: 2 / 2 PASSED

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| Suite Name:         | Assignment5 Tests                                             |
| Test Setup Hook:    | Open Browser And Login (Passed on 2/2 tests)                  |
| Test Teardown Hook: | Logout And Close Browser (Passed on 2/2 tests)                |
| Browser Sessions:   | 2 isolated Chrome driver sessions created & safely destroyed. |

## 7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

### Brief Result:

Both test cases ('TC\_01' and 'TC\_02') passed successfully. For each test case, Robot Framework automatically launched Chrome browser, navigated to the application, logged in, executed the test assertions, performed user logout via the navigation drawer, and closed the browser instance.

### Observation:

- Using Test Setup and Test Teardown at the suite level ensures that every test case begins in a fresh, independent browser context.
- Even if a test step inside a test case fails, Robot Framework guarantees execution of the Test Teardown block, ensuring orphaned browser windows are closed and system resources are properly freed.
- Separating setup and teardown into custom user keywords reduces code redundancy across individual test definitions.

### Conclusion:

In this lab experiment, I successfully automated the test lifecycle management using Robot Framework's Test Setup and Test Teardown features. This approach provides reliable test isolation, prevents state contamination across test runs, and optimizes test suite maintainability.